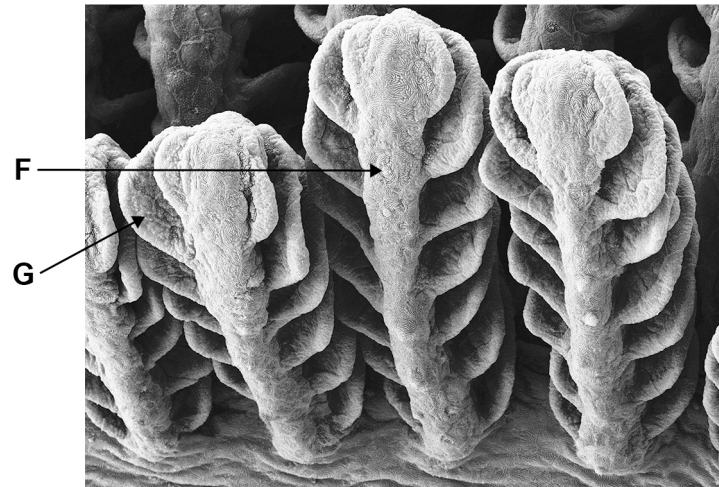


0 6

Figure 4 is an image of a fish gill taken using a scanning electron microscope.

Figure 4



0 6

1

Identify structures labelled **F** and **G**.

[1 mark]

F _____

G _____

0 6

2

Describe and explain the advantage of the counter-current principle in gas exchange across a fish gill.

[3 marks]

Question 6 continues on the next page

Turn over ►



Scientists captured a large number of three species of fish.

The scientists measured:

- the water depth where fish were caught
- the concentration of dissolved oxygen at all water depths
- the mean gill surface area and mean body mass of each species of fish caught.

The scientists calculated the ratio of gill surface area to fish body mass.

The results are shown in **Figure 5** and **Table 4**.

Figure 5

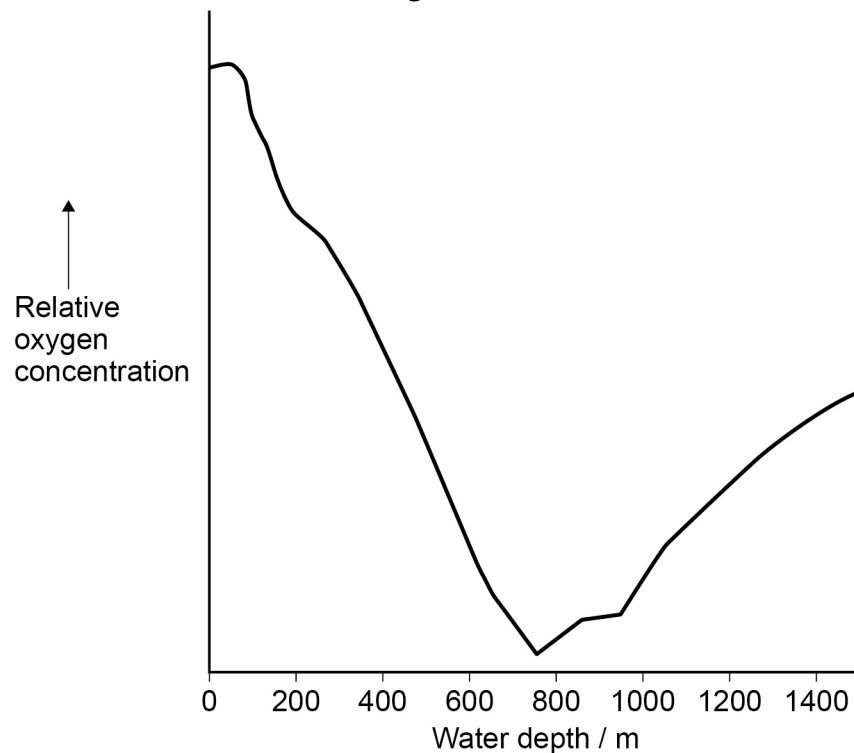


Table 4

Fish species	Water depth where fish were caught / m	Ratio of gill surface area to body mass
<i>Anoplopoma fimbria</i>	450–1280	148:1
<i>Careproctus melanurus</i>	850–1307	124:1
<i>Embassichtys bathybius</i>	1205–1307	20:1

0 6 . 3

Use information in **Figure 5** and **Table 4** to describe the relationship between:

1. dissolved oxygen concentration and increasing water depth
2. dissolved oxygen concentration and the distribution of *Anoplopoma fimbria*.

[2 marks]

1. dissolved oxygen concentration and increasing water depth

2. dissolved oxygen concentration and the distribution of *Anoplopoma fimbria*

0 6 . 4

Using information from **Figure 5** and **Table 4**, what can you conclude about the adaptation of the gas exchange surfaces of these species of fish?**[2 marks]**

8

Turn over for the next question**Turn over ►**